

Data Structures and Algorithms – Sorting Solutions (Python)

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# Import dataset and extract unique alcohol values
import pandas as pd
import random
import time

df = pd.read_csv("winequality-red.csv", sep=';')
alcohol = list(set(df['alcohol']))

# 1. Bubble Sort
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr

def optimized_bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        swapped = False
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True
        if not swapped:
            break
    return arr

# 2. Insertion Sort
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and key < arr[j]:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr

# 3. Merge Sort
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr)//2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    merged = []
    while left and right:
        if left[0] <= right[0]:
            merged.append(left.pop(0))
        else:
            merged.append(right.pop(0))
    merged += left + right
    return merged

# 4. Quick Sort (with multiple pivot strategies)
def quick_sort(arr, pivot_strategy='first'):
    if len(arr) <= 1:
        return arr

    if pivot_strategy == 'first':
        pivot = arr[0]
    elif pivot_strategy == 'last':
        pivot = arr[-1]
    elif pivot_strategy == 'random':
        pivot = random.choice(arr)
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else:
    raise ValueError("Invalid pivot strategy")

less = [x for x in arr if x < pivot]
equal = [x for x in arr if x == pivot]
greater = [x for x in arr if x > pivot]

return quick_sort(less, pivot_strategy) + equal + quick_sort(greater, pivot_strategy)
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